

Jeonghyeon Kim

M.S. Student in Mechanical Engineering, Korea University

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EDUCATION

- **Korea University** *Mar. 2025 – Feb. 2027 (Expected)*
M.S. in Mechanical Engineering Seoul, Republic of Korea
 - Advisor: Prof. Shinsuk Park
 - GPA: 3.78/4.0
 - Research interests: control-oriented dynamics modeling and whole-body control
- **Korea University** *Mar. 2019 – Feb. 2025*
B.S. in Mechanical Engineering Seoul, Republic of Korea
 - GPA: 3.48/4.0

RESEARCH EXPERIENCE

- **Human-Machine Systems Lab, Korea University** *Mar. 2024 – Present*
Graduate Researcher Seoul, Republic of Korea
Balance-Aware Whole-Body Control for Humanoid Standing Manipulation
 - Formulated a **Balance Demand Descriptor (BDD)** based on centroidal dynamics to represent lower-body balance demands induced by upper-body motion, payloads, and external interaction.
 - Integrated BDD into a reinforcement-learning locomotion policy to distinguish demands manageable within the current stance from those requiring corrective stepping.
 - Designed simulation evaluations involving payload variation and continuous upper-body disturbances to assess physical validity and behavioral utility.
- **Visual Perception for Quadrupedal Object Interaction**
 - Designed a dual-camera perception pipeline combining global workspace observation and local near-field sensing in a common coordinate frame.
 - Enabled task-specific perception under viewpoint changes and partial occlusion during quadrupedal locomotion.
- **Real-Time Architecture for a Generative-AI-Based Mobile Manipulator**
 - Designed the communication architecture connecting a simulation environment, an LLM, a user interface, and a physical mobile manipulator.
 - Reorganized asynchronous data exchange around the robot's control cycle to prevent cross-module communication latency from interfering with real-time control.

PUBLICATIONS & MANUSCRIPTS

Submitted Manuscripts

- H.-Y. Ju, **J.-H. Kim**, and S.-S. Park, "A Study on Improving Sim-to-Real Accuracy through Dynamics Parameter Tuning in Reinforcement Learning-based Mapless Navigation," *Journal of Institute of Control, Robotics and Systems*, submitted, Sep. 2026.

Manuscripts in Preparation

- **J.-H. Kim** and S.-S. Park, "Balance Demand Descriptor: A Physics-Informed Representation of Whole-Body Dynamics for Humanoid Standing Manipulation," *manuscript in preparation*.

Conference Presentations

- **J.-H. Kim** and S.-S. Park, "Obstacle Detection Using ORB-SLAM and UV-Disparity with a Low-Cost Quadruped Robot," *KSPE 2025 Spring Conference*, Jeju, Republic of Korea, May 2025.

SELECTED PROJECTS

- **Quadruped Robot Obstacle Avoidance — Team Leader** *Mar. 2024 – Jan. 2025*
 - Developed an obstacle detection and avoidance pipeline for the DOGZILLA quadruped robot using ORB-SLAM and UV-disparity.
- **Ladder-Crossing Robot — Team Leader** *Mar. 2023 – Sep. 2023*
 - Designed a ladder-crossing mechanism and implemented an Arduino-based control system.
- **Solar-Tracking Parasol — Team Leader** *Sep. 2023 – Nov. 2023*
 - Designed the linkage mechanism and implemented a sun-tracking control system.

HONORS AND AWARDS

- **Master's Research Encouragement Grant**, National Research Foundation of Korea *2025*
- **President's Award**, Creative Challenger Program (CCP), Korea University *Jan. 2025*
- **Bronze Award**, Creative Comprehensive Design Competition, Korea University *Sep. 2023*
- **Design Award**, Capstone Design Fair, Korea University *Nov. 2023*

TEACHING EXPERIENCE

Teaching Assistant, Korea University, Seoul, Republic of Korea

- **Robotics** *Spring 2026*
- **Computer-Aided Design** *Fall 2025*
- **Dynamics, English-Taught Course** *Summer 2025*
- **Kinematics of Mechanisms** *Spring 2025*

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB
Robotics: ROS2, Pinocchio, URDF/XACRO
Simulation: Isaac Gym, MuJoCo, Gazebo
Learning & Optimization: Reinforcement Learning, Trajectory Optimization
Perception: OpenCV
Engineering Tools: Git, Creo, NX, SolidWorks, Arduino
Operating Systems: Linux (Ubuntu)